EE3032 Introduction to VLSI Design—Homework 2

1. Fig. 1(a) shows an inverter driving a capacitive load C_L. If two input stimuli as shown in Fig. 1(b) are applied, the fall time of the output Vout for the input stimulus a is shorter than that for the input stimulus b. Why? Explain the reasons. (3 points)

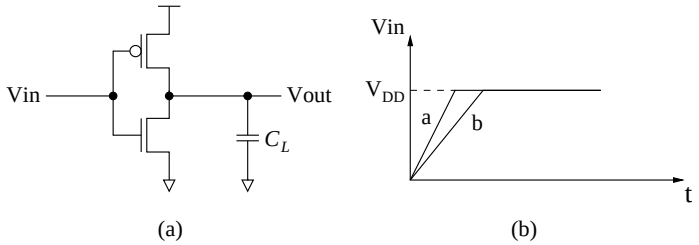


Figure 1: (a) An inverter drives a load C_L. (b) Two input waveforms with different rise times.

2. A MOS transistor can be modeled as a π-model. For example, Fig. 2(a) shows the π-model of an NMOS transistor, which consists of a turn-on equivalent resistance R and two diffusion capacitance C_d. Derive the relationship between the number of inputs (fanin) and the t_r and t_f of the x-input NOR gate without loading. (3 points)

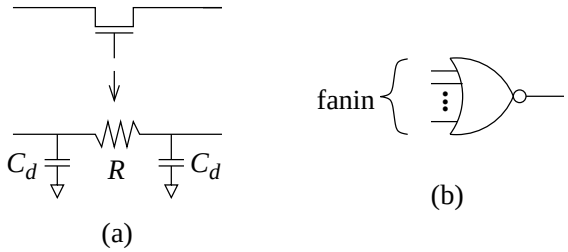
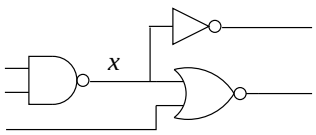


Figure 2: (a) MOS transistor and its RC model. (b) A x-input NOR gate.

3. Assume that the size of the two transistors of an inverter is as follows: M1=$W_p/L_p = 4$ and M2=$W_n/L_n = 2$. If Vin=$\frac{1}{2}V_{DD}$, the Vout=$\frac{1}{2}V_{DD}$. Calculate the ratio of the μ_n to μ_p. (2 points)

4. Fig. 4 shows a simple logic circuit. Assume that the gate capacitance of the invert is C_g; the β value of the PMOS and NMOS of the inverter is β_p and β_n, respectively. Also, $\beta_n = \beta_p$. If the equivalent β of PMOS network and NMOS network of the two-input NOR gate is β_{peq} and β_{neq}, respectively. Assume that $\beta_{peq} = \beta_p$ and $\beta_{neq} = \beta_n$; $\mu_n = 2\mu_p$. Calculate the capactiance of the node x without considering the parasitic capacitances of wire and drain. (2 points)